

TrialGPT:

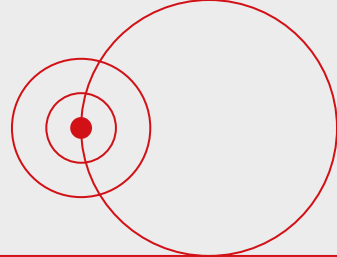
End-to-end patient-to-trial matching with LLMs

Jin et al. (2024), Nature Communications 15: 53081

Nathan Gin

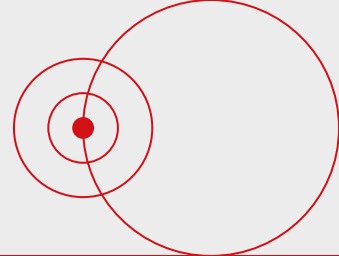
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Motivation: Why This Matters



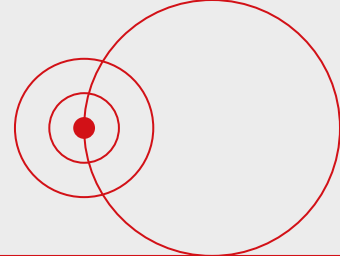
- Eligibility criteria are long, complex, and unstructured
- Manual screening is slow (weeks) and inconsistent
- Recruitment delays lead to underpowered trials & protocol amendments
- Eligibility errors threaten internal validity even if randomization is perfect

Recruitment Bottleneck in Clinical Trials

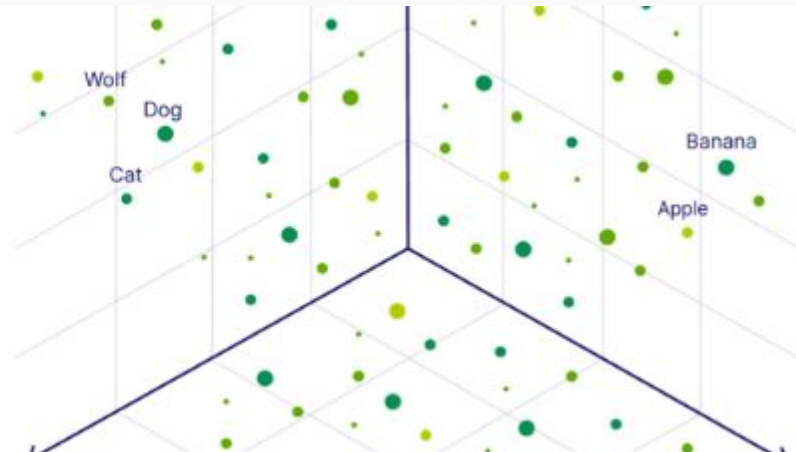
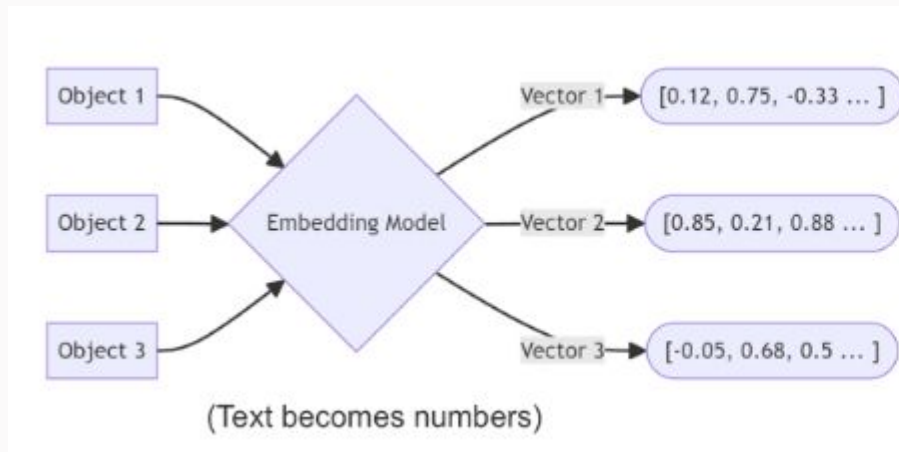


- ~80% of trials fail to meet enrollment timelines
- ~25–30% of cancer trials fail due to insufficient enrollment
- Manual chart review is labor-intensive and varies by coordinator
- Eligibility misclassification introduces selection bias

Why Use AI for Trial Matching?

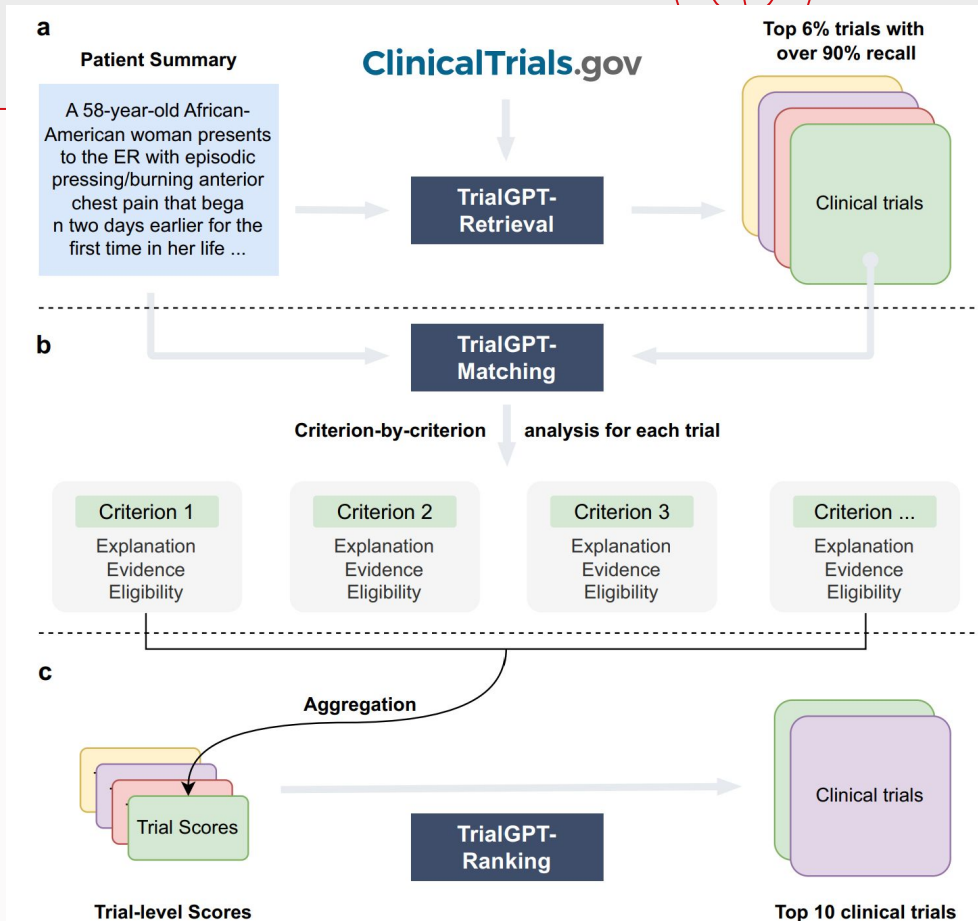


- Need automation for scale (> 20,000 active trials in U.S.)
- Traditional keyword systems miss context (e.g., “high blood sugar” ≠ “hyperglycemia”)
- LLMs capture synonyms, reasoning, and cross-criterion logic
- LLMs reason over full patient notes lead to better recall, fewer missed eligible patients
- LLMs can explain predictions and highlight relevant evidence sentences
- More consistent screening leads to reduced selection bias and better external validity

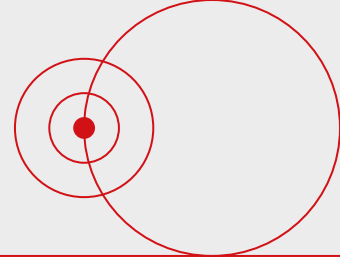


TrialGPT Architecture and Modules

- 3-stage pipeline:
 - **Stage 1: Retrieval** — Narrow down ~20,000 trials to a candidate set
 - **Stage 2: Criterion-level Matching** — Evaluate patient note against each inclusion/exclusion rule
 - **Stage 3: Trial-level Ranking** — Aggregate scores (inclusion/exclusion/relevance) to produce final ranking
- Uses structured + chain-of-thought prompting to generate explanations
- Uses GPT-4 for reasoning over clinical text



Evaluation Data and Study Design



- Public benchmark cohorts from prior patient-trial retrieval tasks:
 - SIGIR 2016 (n = 58, short clinical notes)
 - TREC 2021 CT (n = 75, long oncology notes)
 - TREC 2022 CT (n = 50, high-complexity trials)
- ~75,000 patient-trial pairs manually labeled for eligibility: eligible, excluded, or irrelevant
- These datasets simulate real clinical screening: each patient must be compared against hundreds of candidate trials.

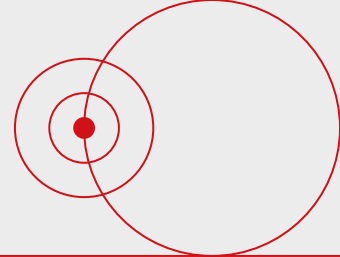
Table 1 | Baseline statistics of the three patient cohorts used in this work

Cohort	SIGIR	TREC 2021 CT	TREC 2022 CT
N	58	75	50
Age (year)	38.5 ± 23.7	41.6 ± 19.4	35.3 ± 20.2
Sex (male: female)	29: 29	38: 37	28: 22
Note length (words)	88.7 ± 36.8	156.2 ± 45.4	109.9 ± 21.6
Eligible trials/patient	7.3 ± 6.7	74.3 ± 49.0	78.8 ± 67.3
Potential trials/patient	11.7 ± 10.2	None	None
Excluded trials/patient	None	80.3 ± 60.3	60.7 ± 65.5
Irrelevant trials/patient	47.1 ± 19.5	323.2 ± 93.2	568.4 ± 164.1
Considered initial trials	3621	26149	26581

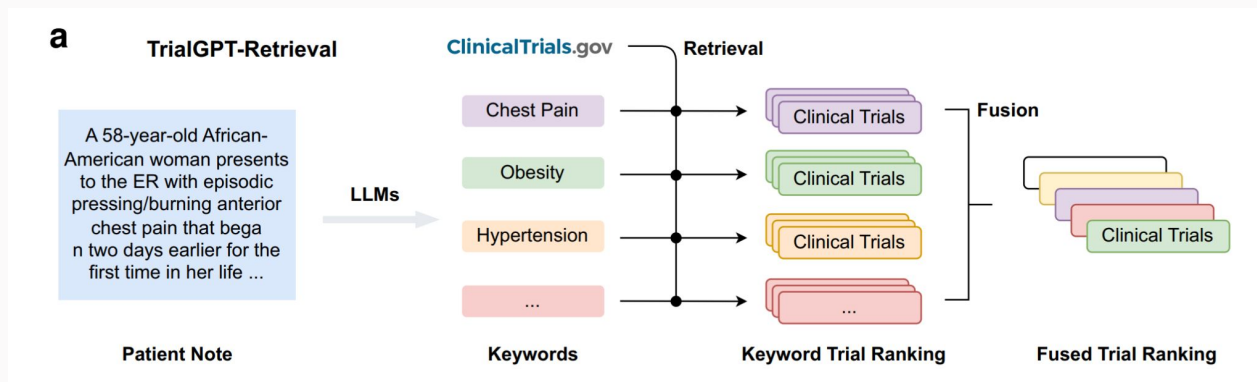
We show the mean ± standard deviation for applicable variables. “None” denotes there is no such eligibility label in the corresponding cohort. SIGIR: the patient-trial matching cohort published at the Special Interest Group on Information Retrieval (SIGIR).

TREC the Text REtrieval Conference (TREC), CT the clinical trials track at TREC.

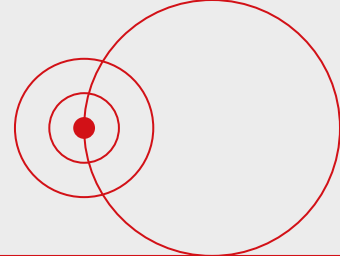
TrialGPT-Retrieval: Finding Relevant Trials



- LLM generates keywords from patient note (up to 32)
- Hybrid retrieval = BM25 (lexical) + MedCPT (semantic) → Reciprocal Rank Fusion
- Narrows from ~20,000 trials to ~1,000 candidates
- Only 5.5% of database needed to capture 90% of relevant trials
- High recall ensures relevant trials are not lost

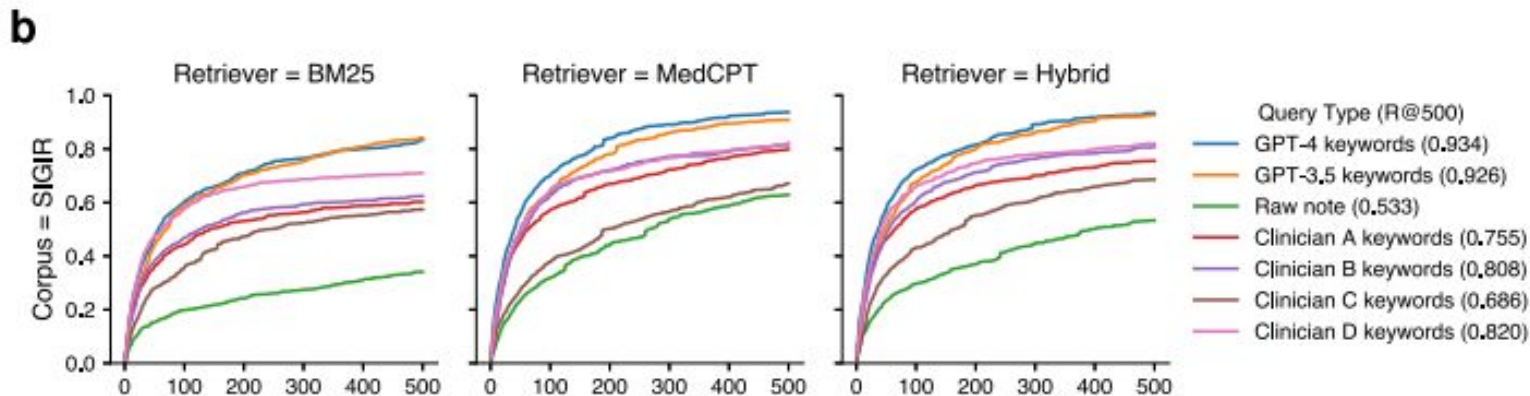


TrialGPT-Retrieval: Results

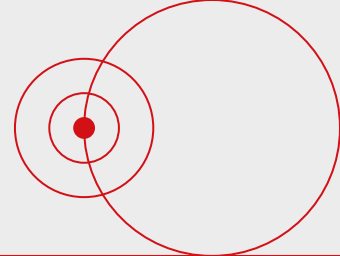


- GPT-4 keywords outperform clinicians & GPT-3.5
 - 93% recall at depth 500
- Reciprocal Rank Fusion (RRF) combines BM25 + MedCPT
- Ensures matching only analyzes likely-relevant trials

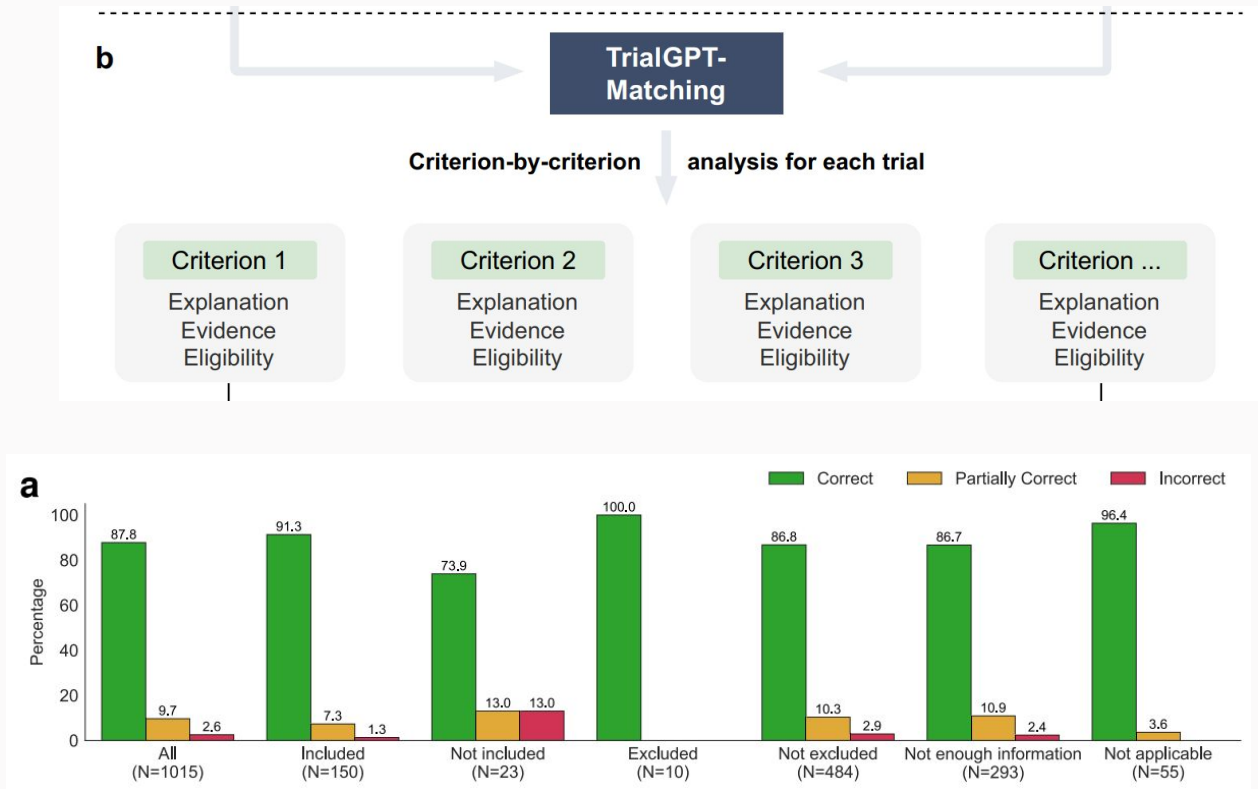
$$s_j = \sum_{\text{Ret}} \sum_{i=1}^K \frac{1}{i \times (\text{Rank}(\text{Ret}, w_i, t_j) + C)}$$



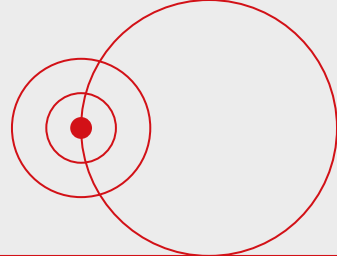
TrialGPT-Matching: Criterion-Level Eligibility



- LLM reads patient note + 1 criterion at a time
- Produces:
 - Eligibility label
 - Evidence sentences
 - Explanation (chain-of-thought)
- Helps clinicians audit correctness



How Criterion Labels Are Produced



- TrialGPT uses a cross-encoder that reads patient note + one eligibility criterion at a time
- Model predicts NLI labels: entailment, contradiction, neutral
- Map NLI outputs to eligibility categories:
 - Included = entailment
 - Not included = contradiction
 - No information = neutral
 - Excluded / not-excluded handled similarly for exclusion criteria
- Aggregation computes:
 - % inclusion criteria met
 - % exclusion criteria triggered
 - General relevance score (0–100)
 - Eligibility score (0–100)

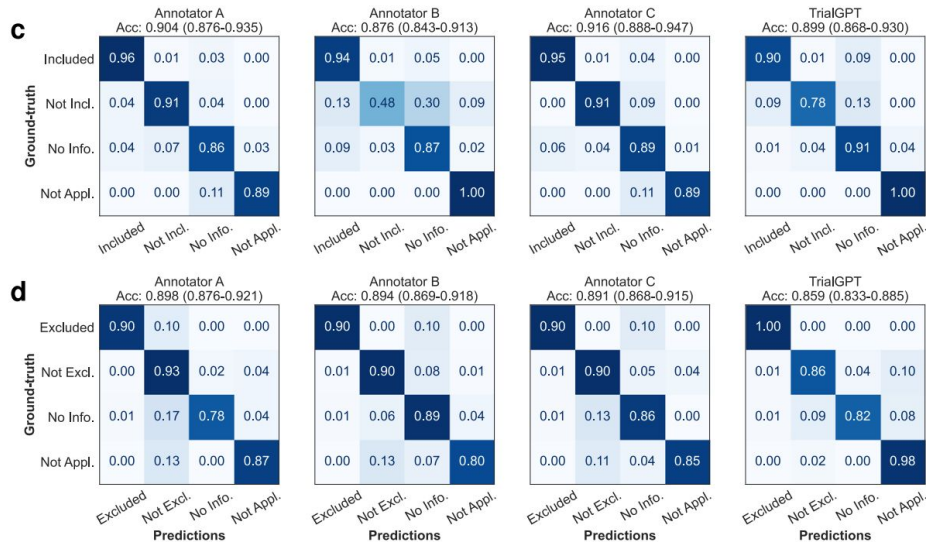
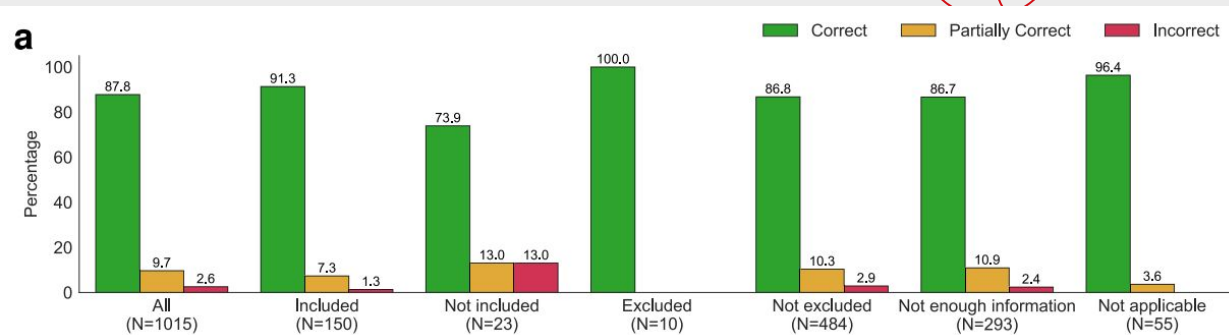
$$\% \text{ met inclusion criteria} = \frac{|\{E(i_x) = \text{included} \mid x = 1, 2, \dots, M\}|}{M'}$$

$$\% \text{ met exclusion criteria} = \frac{|\{E(e_y) = \text{excluded} \mid y = 1, 2, \dots, N\}|}{N'}$$

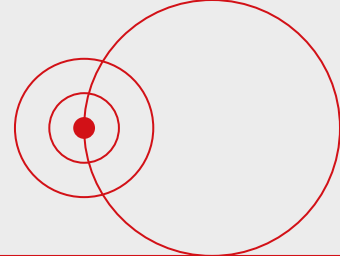
Evaluation of Criterion-Level Predictions



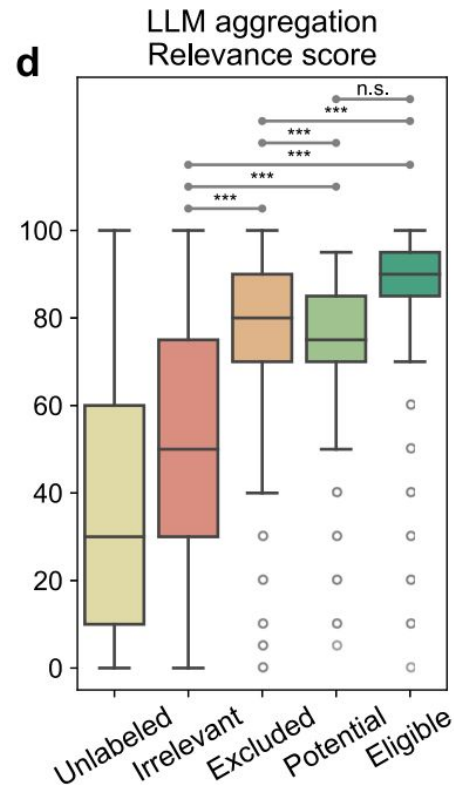
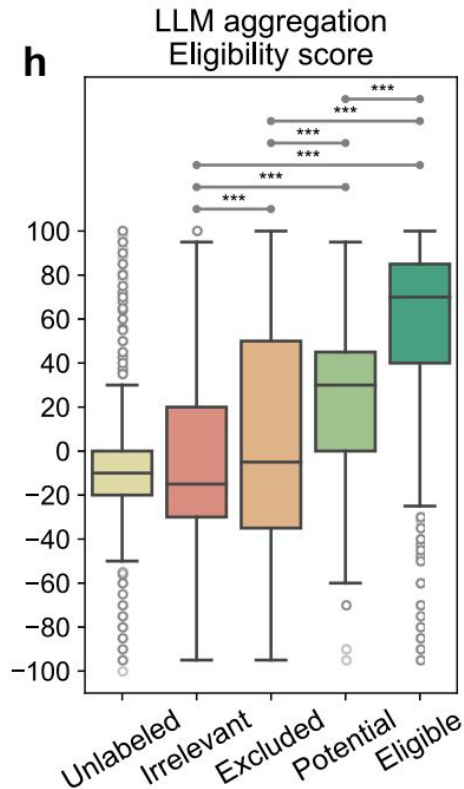
- 1,015 patient-criterion pairs manually reviewed
- ~90% correctness for explanations
- Eligibility accuracy: ~87–89%
- Evidence extraction F1 $\approx$ 0.90
- Confusion matrices show:
 - Strong diagonal
 - Most errors come from unclear EHR text or abbreviations
 - Performance comparable to human annotators



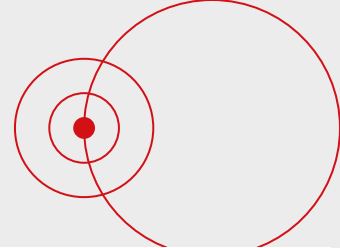
TrialGPT-Ranking: Trial-Level Aggregation



- Aggregates criterion-level outputs into a single trial score
- Uses two aggregation strategies:
 - **Linear:** % inclusion criteria met, % exclusion criteria triggered
 - **LLM-based:** relevance score + eligibility score (0–100)
- High-scoring pairs meet most inclusion criteria and few exclusion criteria
- Produces a ranked list of best-matching trials
- Aggregated scores separate trial–patient pairs into:
 - **Irrelevant**
 - **Potential**
 - **Eligible**
- LLM aggregation outperforms linear scoring



How Well Does It Work?



TrialGPT (GPT-4) vs best baseline
(BioLinkBERT, MedNLI-trained):

- NDCG@10: 0.73 vs 0.48
- P@10: 0.67 vs 0.43
- AUROC: 0.80 vs 0.62
- ≈44% improvement in overall ranking & exclusion accuracy
- LLM aggregation (relevance/eligibility scores) outperforms linear aggregation and all encoder-decoder baselines
- Strongest performance across all evaluation settings (SIGIR, TREC21, TREC22)

Table 2 | Performance of different methods for ranking and excluding clinical trials

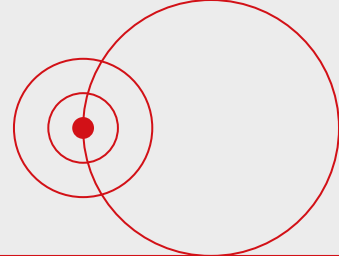
Application			Ranking		Excluding	Overall
Method / Metric			NDCG@10	P@10	AUROC	Average
SciFive ⁴⁶ (encoder-decoder)	Further trained on MedNLI ³⁷		0.4271	0.3787	0.5895	0.4652
BioBERT ⁴⁷ (dual-encoder)	Further trained ⁴⁸ on MNLI ⁴⁹ , SNLI ⁵⁰ , SciNLI ⁵¹ , SciTail ⁵² , MedNLI ³⁷ , and STSB ⁵³		0.4091	0.3746	0.5952	0.4596
PubMedBERT ⁵⁴ (dual-encoder)			0.4327	0.3874	0.5976	0.4726
SapBERT ⁵⁵ (dual-encoder)			0.4148	0.3738	0.5933	0.4606
BioLinkBERT ³⁶ (cross-encoder)	Further trained on MedNLI ³⁷		0.4797	0.4281	0.6176	0.5085
TrialGPT-Ranking (GPT-3.5)	Feature combination		0.5395	0.5115	0.6582	0.5697
TrialGPT-Ranking (GPT-4)	Linear aggr.	Sign(% Included)	0.6097	0.5653	0.6765	0.6172
		Sign(% Not included)	0.4923	0.4556	0.6652	0.5377
		Sign(% Excluded)	0.3930	0.3715	0.6477	0.4707
		Sign(% Not excluded)	0.4130	0.3886	0.5820	0.4612
	LLM aggr.	Sign(Relevance)	0.7281	0.6700	0.7402	0.7128
		Sign(Eligibility)	0.7252	0.6724	0.7895	0.7290
	Feature combination		0.7275	0.6688	0.7979	0.7314

The Sign() function assigns suitable signs for the corresponding task, e.g., for “% Included”, it will be “+” for ranking and “-” for excluding clinical trials. Aggr.: aggregation. NDCG@10: normalized discounted cumulative gain at 10. P@10: precision at 10.

AUROC the area under the receiver operating characteristic curve.

$$\text{NDCG}@k = \frac{\text{DCG}@k}{\text{IDCG}@l}$$

Real-World Test: Physicians with TrialGPT

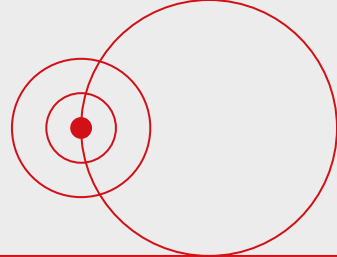


- 6 trials × 6 patients → 36 screening tasks
- Two physicians: half of tasks with TrialGPT, half without
- Decision accuracy: 97% vs 92% (difference not statistically significant)
- Screening time reduced by 42.6% on average
 - Greater savings for longer, more complex cases
- Same decisions with or without AI (97% concordance)
- P-values show time reduction is statistically significant (paired t-test)

	Short Cases	Long Cases	Annotator X	Annotator Y	All (36)
Without TrialGPT (18)	42.7 s	80.2 s	65.4 s	57.5 s	61.5 s
With TrialGPT (18)	25.2 s	45.3 s	37.8 s	32.7 s	35.3 s
Time Saving	41.0%	43.5%	42.2%	43.1%	42.6%
P-value	0.0182 [†]	0.0003 [†]	0.1633 [‡]	0.2472 [‡]	1.75e-5 [†]

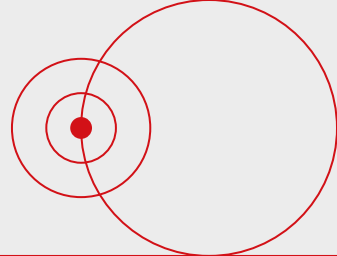
[†]: Paired t-test; [‡]: independent t-test

Randomization & Blinding



- TrialGPT is used before randomization
- Standardizes eligibility assessment → stronger internal validity
- Blinding preserved (AI sees no treatment-arm info)
- Reduces systematic eligibility bias across sites
- Eliminates coordinator-to-coordinator variability

Statistical Considerations



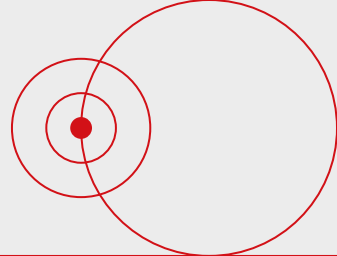
Better screening → more accurate:

- Event-rate assumptions
- Dropout projections
- Screen-failure rates

Other benefits:

- Reduces eligibility misclassification error
- Improves power calculations (fewer screen failures)
- Criterion-level outputs allow sensitivity analyses (e.g., impact of mislabeling one criterion)

Bias and Limitations



Data limitations

- Synthetic cohorts → uncertain external validity
- Domain shift across institutions
- Demographic under-representation

Model limitations

- Dependence on GPT-4
- Hallucination risk
- Knowledge gaps in rare conditions

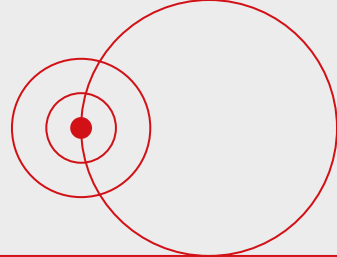
Annotation & documentation issues

- Ambiguous criteria → noisy labels
- Variability in EHR documentation

Ethical considerations

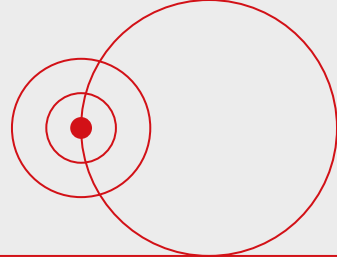
- Transparency & explainability
- Bias propagation
- Need for human oversight

Practical Impact & Future Directions



- Faster enrollment → fewer protocol deviations
- Standardized eligibility screening across sites
- Enhances support for adaptive and platform trials
- Potential integration with:
 - Multimodal inputs (EHR + imaging + labs)
 - Prospective clinical evaluation
 - Smaller, domain-specific medical LLMs
 - Workflow automation in trial centers

References



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